AR-Mini · A team of three and six AI agents in one home lab

Teddy designs and decides. AI agents propose. Mom and Dad close the loop in the physical world. · Sofia · 07.10 → 01.11.2026

How Teddy got here

from the journal, grade 1 → 12

1 · sport
IRONMAN spark
Triathlon and ultra trail: wanting to see how far she can go. Water became personal.

2 · idea
Dreaming of robots
What if a machine could reach a swimmer in trouble faster than a person?

3 · 9th grade
CS50 + C++
Harvard CS50 and C++ in the Software & Hardware class at school.

4 · from 09.2025
1 year of research
AmphiReach: theory and simulation (ROS 2, Gazebo). Paper accepted by NHSJS.

5 · conference
First physical thing
A small Lego build for the conference at the Bulgarian Academy of Sciences (IMI).

6 · summer 2026
Stanford labs
CS106A and ENGR 40M: labs and Arduino projects, hands on real circuits.

7 · 2026
Home lab
3D printer, bench tools, a workspace on the floor at home.

8 · NOW
Real prototype
AR-Mini, a ~1:9 amphibious rescue robot. 26 days to 01.11, with AI agents and family.

TEDDY · student · system architect

Sets the requirements · writes the hypothesis · reviews every AI proposal · signs off every version · owns the commits

decides what to build · decides what is true · designs the CAD · wires and solders · owns the repo

requirements, hypothesis, photos ↓ ↑ *proposals: CAD, wiring, code, orders*

what to print, test, film ↓ ↑ *results*

AI agents · six roles, each with its own context

Claude Projects (one per role) + Claude Code on the laptop for firmware. Agents propose; they never mark anything as tested.

Coordinator
Claude Project

- reads the loop tracker every morning
- 3 priorities for today
- writes tasks for Mom and Dad
- cuts scope when a release slips

Mech
Claude Project



- OpenSCAD model v1.9 →
- small test sections first
- collision checks in 3D
- print file + hypothesis

Elec
Claude Project



- ESP32-S3 DevKit-Lipo, drivers
- fuse tree before first power
- pin-to-pin wiring tables
- what to measure, expected value

Firmware
Claude Code · laptop



- PlatformIO, C++
- compiles and flashes over USB
- CSV over serial at 10 Hz
- Teddy commits, agent suggests

Sourcing
Claude Project



- keeps the BOM
- fast vs cheap option
- delivery time to Sofia
- statuses: ordered → delivered

Docs
Claude Project

git · GitHub

- turns closed loops into portfolio
- FAILS_AND_FIXES log
- only what has a photo or video
- marks human vs AI work

Home lab · the physical world

Everyone lives where the printer and the bench are.

Mom · Tsvety
operator · hands and camera

Dad · Dimi
operator · hands and camera

What they do (tasks come from the Coordinator)

Printer	load the file, photo of first layer and of the part next to a ruler
Parcels	photo of label and contents → Sourcing updates the BOM
Tests	run it, read the meter, photo of every number
Video	landscape, 5 s before and after, no music
Safety	smoke, smell or heat: power off first, then photo

They do not design. They are the eyes and hands that make the loop fast.

The loop · logged in the AR-Mini Loop tracker (phone)

The clock starts at "Start" and stops at "Close". Only a person who saw the result can close a loop.

Hypothesis — Teddy: what we expect, with a number → **Proposal** — agent: change to CAD, wiring or code → **Build** — print, wire, flash → **Measure** — Mom, Dad or Teddy: test + photo/video → **Close** — pass / fail + the next fix

next version · goal: hours per loop, not days

Why it is lean

3 people, 1 home lab

6 AI agent roles

0 shipping between team members

Releases (planned)

R1 · 13.10	section floats, draft measured, arm lifts Pesho in the bathtub
R2 · 22.10	full hull, thruster mount, swims in the pool
R3 · 29.10	heat-seek → approach → lift
01.11	video + MIT Maker Portfolio

Rules that keep it honest

- AI proposes, a measurement decides.
- No invented sizes, pins or specs: unknown means a measuring task.
- Portfolio numbers come only from the tracker.
- Design and decisions are Teddy's; parents help with hands and camera.